

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

School of Computing — Department of Computer Science & Engineering

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Course: 21CSC203P — Advanced Programming Practice	Department: Computer Science and Engineering

Unit I | Session 5 | SLO 2: Programming Paradigm Hierarchy (Code: 1052)

Q: 1. Discuss the transition from Unstructured to Structured and Procedural Imperative Programming.

Ans: Discussion: Early imperative languages relied on unconditional jumps (goto), resulting in fragmented codebases with high maintenance overhead. The structured programming paradigm solved this by enforcing modular single-entry, single-exit constructs (sequence, selection, iteration). Procedural programming further refined this by abstracting repetitive code blocks into parameterized procedures and functions with local scoping.

Q: 2. Provide a Java code example illustrating the imperative paradigm with state mutation.

Ans: Java Implementation:

```
public class ImperativeStateDemo {
    public static void main(String[] args) {
        int total = 0; // Initial state
        int[] numbers = {10, 20, 30, 40};
        for (int i = 0; i < numbers.length; i++) {
            total += numbers[i]; // State mutated step-by-step
        }
        System.out.println("Cumulative Total: " + total);
    }
}
```